Assignment - 5

Code -

```
#include <iostream>
#include <vector>
using namespace std;

int N;

bool isSafe(vector<vector<int>>& board, int row, int col) {
    for (int i = 0; i < row; i++)
        if (board[i][col] == 1)
            return false;

    for (int i = row - 1, j = col - 1; i >= 0 && j >= 0; i--, j--)
        if (board[i][j] == 1)
            return false;

    for (int i = row - 1, j = col + 1; i >= 0 && j < N; i--, j++)
        if (board[i][j] == 1)
            return false;

    return true;
}

bool solveNQueens(vector<vector<int>>& board, int row) {
    if (row == N)
        return true;

    for (int col = 0; col < N; col++) {
        if (isSafe(board, row, col)) {
            board[row][col] = 1;

            if (solveNQueens(board, row + 1))
                return true;
```